

# ***NLP***

## Word Embeddings

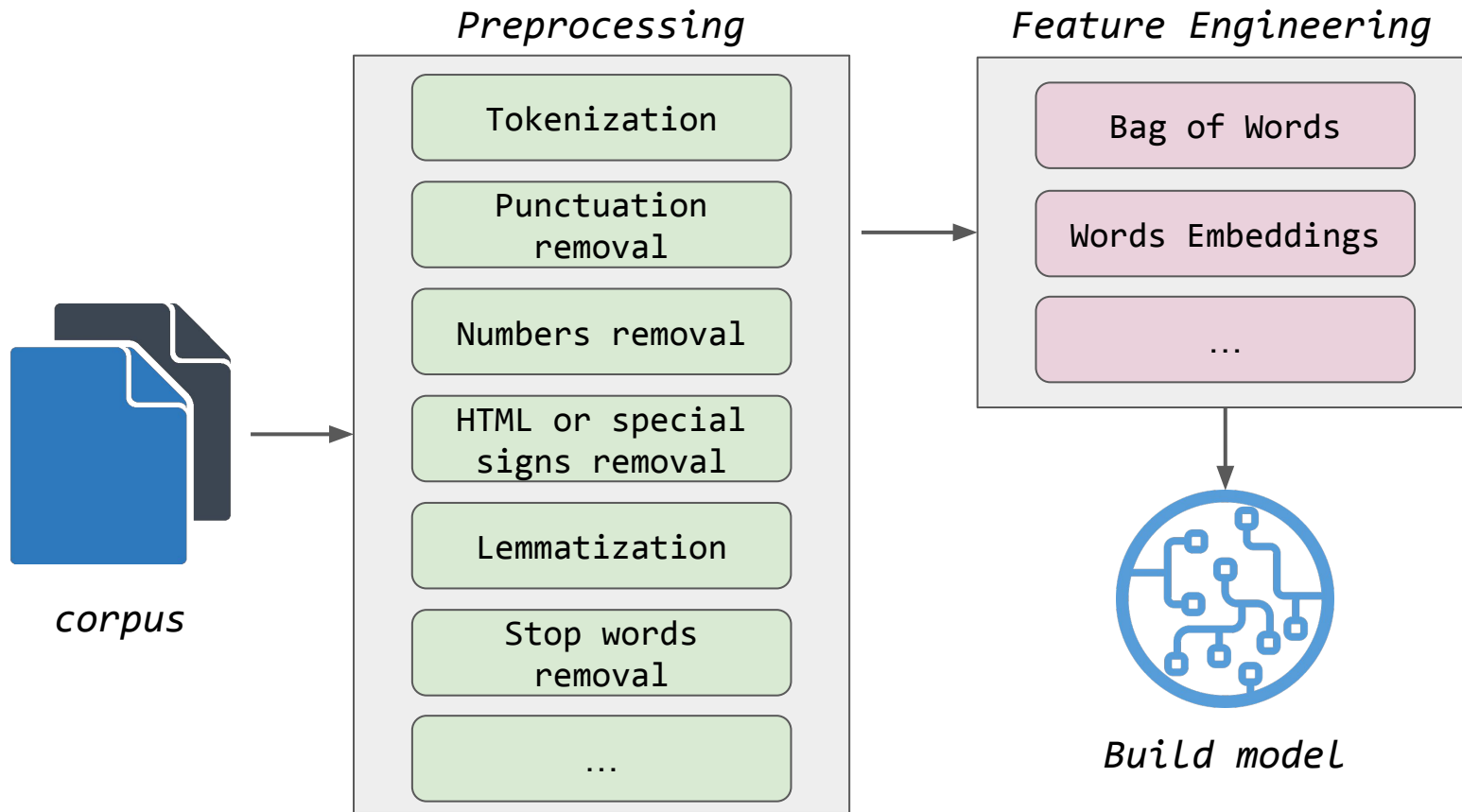
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# Preprocesamiento de texto

[LINK GLOSARIO](#)



# TF-IDF (Term frequency-Inverse document frequency)



Se utiliza como indicador de cuán importante es una palabra (término) en un documento.

$$\text{TF-IDF}_{(n,d)} = \text{TF}_{(n,d)} \times \text{IDF}_{(n)}$$

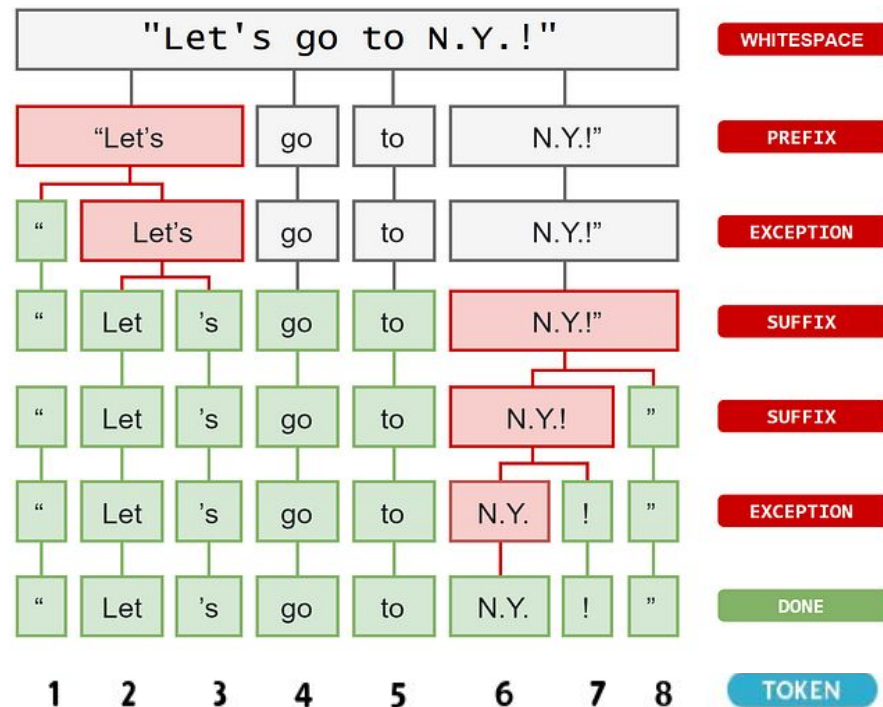
Peso de un término (n) en un documento (d)

Frecuencia de aparición de un término (n) en un documento (d)

Factor IDF de un término (n)



Proceso en el cual una oración o documento es segmentado en términos individuales. Una vez finalizada la segmentación cada término único es referenciado mediante un token.



# Problemas con CountVectorizer/OHE/TF-IDF



*Textos de significado similar pueden ser “ortogonales”*

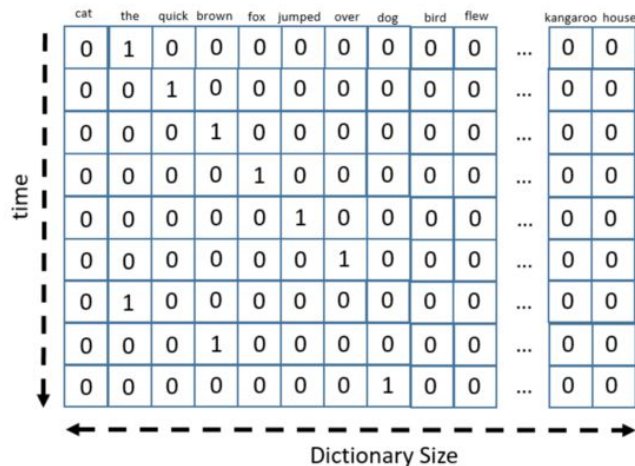
Estoy viajando en colectivo

Voy arriba del bus

Ortogonal -> Similaridad coseno = 0

*La dimensión de los vectores depende del tamaño del vocabulario*

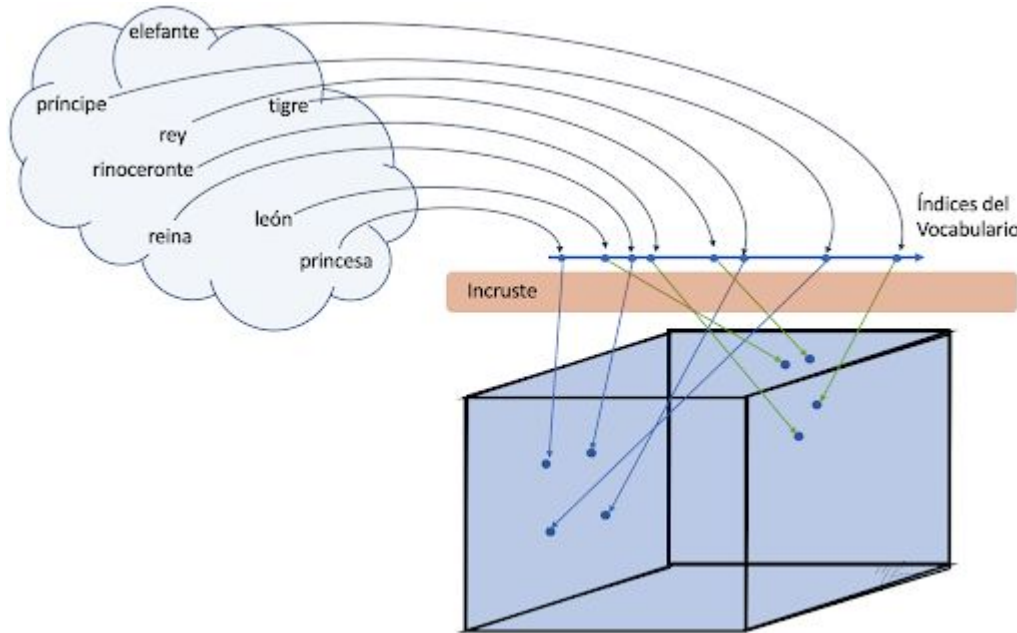
*No aprovechamos la dimensionalidad*



# Embeddings



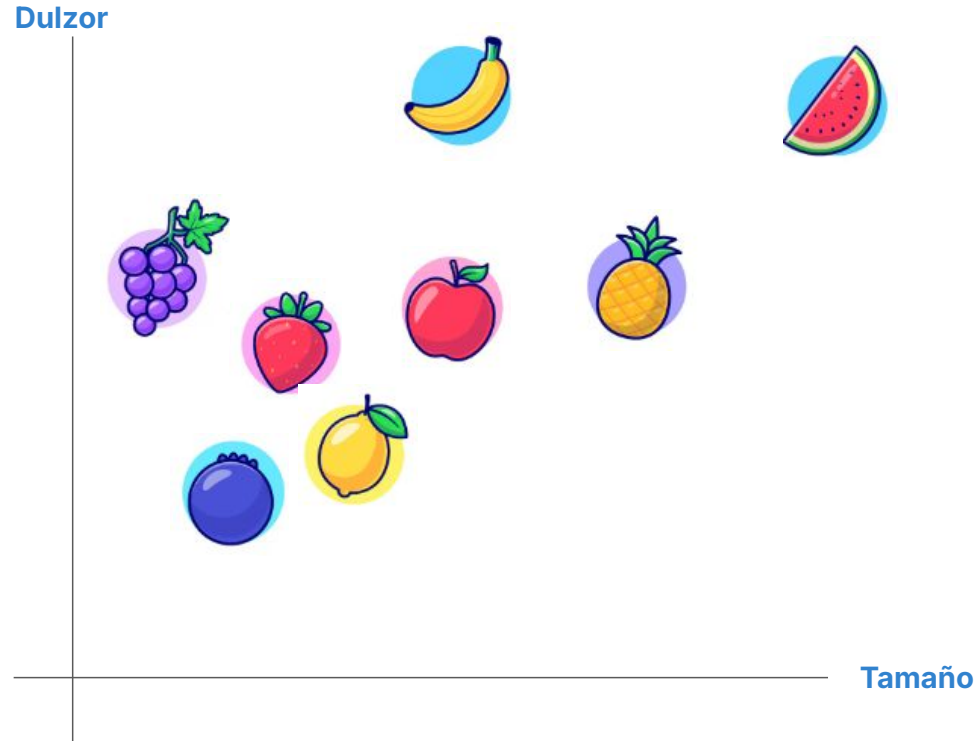
*Un embedding es una representación densa de palabras en un espacio vectorial continuo.*



***Propiedades buscadas:***

- 1. Representación compacta***
- 2. Similitud semántica***
- 3. Comprensión de contexto***

# Atributos como dimensiones



# Atributos como dimensiones



|                                                                                   | Dulzor | Tamaño | Rojo | Tropical | Circular |
|-----------------------------------------------------------------------------------|--------|--------|------|----------|----------|
|  | 4      | 3      | 10   | 3        | 3        |
|  | 10     | 10     | 0    | 10       | 6        |
|  | 3      | 1      | 0    | 2        | 10       |
|  | 5      | 5      | 10   | 2        | 7        |
|  | 7      | 7      | 0    | 7        | 2        |



# Evolución de embeddings

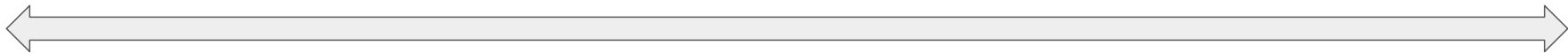


## Word2Vec (2013)

- **Creador:** Google
- **Hito:** Es el momento en que los vectores se vuelven "famosos".
- **Aporte:** Introducen las arquitecturas **CBOW** y **Skip-gram**.

## Fast Text (2016)

- **Creador:** Facebook
- **Hito:** Resuelven el problema de las palabras desconocidas.
- **Aporte:** n-grams de caracteres.



## Glove (2014)

- **Creador:** Universidad de Stanford
- **Hito:** Incorpora estadísticas globales de todo el texto.
- **Aporte:** matriz de co-ocurrencias de todo el corpus.

## ELMo / BERT (2018+)

- **Creador:** Allen Institute (ELMo) y Google (BERT)
- **Hito:** embeddings dinámicos.
- **Aporte:** pasamos de word embeddings a contextual embeddings.



Es una técnica publicada por **Google en 2013** que revolucionó la Inteligencia Artificial. Su gran descubrimiento fue que podíamos enseñarles a las máquinas el **significado** de las palabras simplemente haciéndolas "leer" mucho texto, sin diccionarios.

**Antes:** Las palabras eran etiquetas aisladas (IDs).

**Con Word2Vec:** Las palabras son **vectores** (flechas o coordenadas) en un espacio de muchas dimensiones.

**La hipótesis distributiva:** "Conocerás a una palabra por las compañías que tiene"

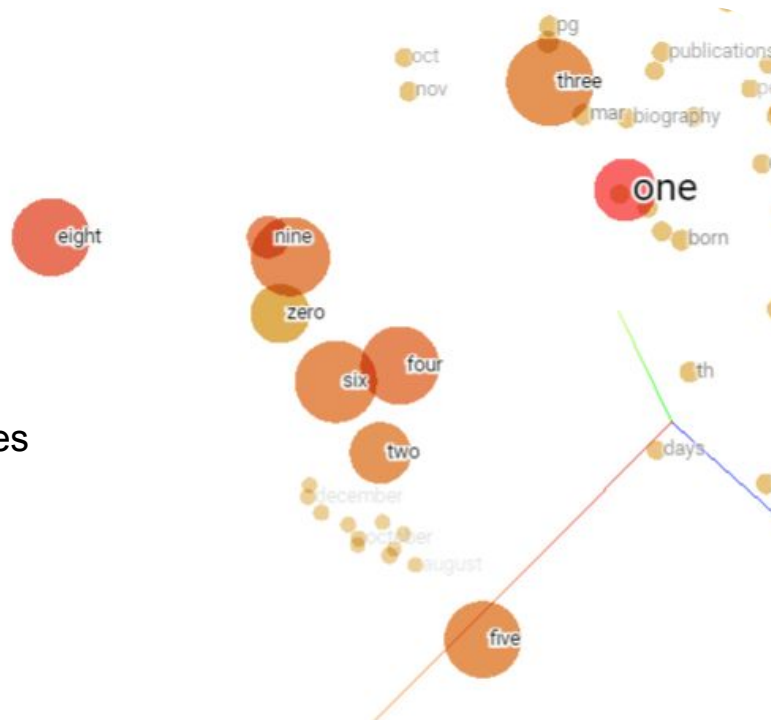
# Word Embeddings



*Las palabras que tienen un significado similar tendrán una representación similar como embeddings*

<http://projector.tensorflow.org/>

Veremos una proyección en 3 dimensiones de los embeddings multidimensionales.



# GloVe y fastText



Embeddings pre-entrenados basados en diferentes enfoques:

## GloVe



Tokenización basada en palabras



Entrenado con textos de Wikipedia, Common Crawl y GigaWord 5



Se basa en calcular la matriz de co-ocurrencia de palabras y estimar el cociente de probabilidad de aparición.

## fastText



Tokenización basada en N-Grams de caracteres (3 a 6). Mejora la interpretación de sufijos y prefijos



Entrenado con una colección de 8 corpus (portales de noticias, reviews, Wikipedia)



Basado en word2vec (CBOW/Skip-Gram)



Puede crear un embedding de una palabra que nunca vió

# Matriz de co-ocurrencia



**Obtención del corpus:** Se reunió las letras de sus canciones desde fuentes en línea, como letras.top y Buenamusica.com.

## Preprocesamiento del texto:

- Convertir todo el texto a minúsculas.
- Eliminar signos de puntuación.
- Tokenizar el texto para dividirlo en palabras individuales.
- Eliminar palabras vacías comunes en español, como "el", "la", "y", etc.

**Cálculo de frecuencias:** Contar la frecuencia de cada palabra y seleccionar las 10 más comunes.

## Construcción de la matriz de co-ocurrencia:

- Definir una ventana de contexto de 5 palabras alrededor de cada término.
- Contar cuántas veces cada par de palabras apareció dentro de la misma ventana.

|         | amor | vida | luz | sol | piel | alma | tiempo | ojos | corazón | cielo |
|---------|------|------|-----|-----|------|------|--------|------|---------|-------|
| amor    | 0    | 15   | 12  | 10  | 8    | 14   | 9      | 11   | 13      | 10    |
| vida    | 15   | 0    | 11  | 9   | 7    | 13   | 8      | 10   | 12      | 9     |
| luz     | 12   | 11   | 0   | 8   | 6    | 10   | 7      | 9    | 11      | 8     |
| sol     | 10   | 9    | 8   | 0   | 5    | 9    | 6      | 8    | 10      | 7     |
| piel    | 8    | 7    | 6   | 5   | 0    | 7    | 4      | 6    | 8       | 5     |
| alma    | 14   | 13   | 10  | 9   | 7    | 0    | 8      | 11   | 13      | 10    |
| tiempo  | 9    | 8    | 7   | 6   | 4    | 8    | 0      | 7    | 9       | 6     |
| ojos    | 11   | 10   | 9   | 8   | 6    | 11   | 7      | 0    | 12      | 9     |
| corazón | 13   | 12   | 11  | 10  | 8    | 13   | 9      | 12   | 0       | 10    |
| cielo   | 10   | 9    | 8   | 7   | 5    | 10   | 6      | 9    | 10      | 0     |

# GloVe y fastText



Embeddings pre-entrenados basados en diferentes enfoques:

## GloVe



Tokenización basada en palabras



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Se basa en calcular la matriz de co-ocurrencia de palabras y estimar el cociente de probabilidad de aparición.

## fastText



Tokenización basada en N-Grams de caracteres (3 a 6). Mejora la interpretación de sufijos y prefijos



Entrenado con una colección de 8 corpus (portales de noticias, reviews, Wikipedia)



Basado en word2vec (CBOW/Skip-Gram)

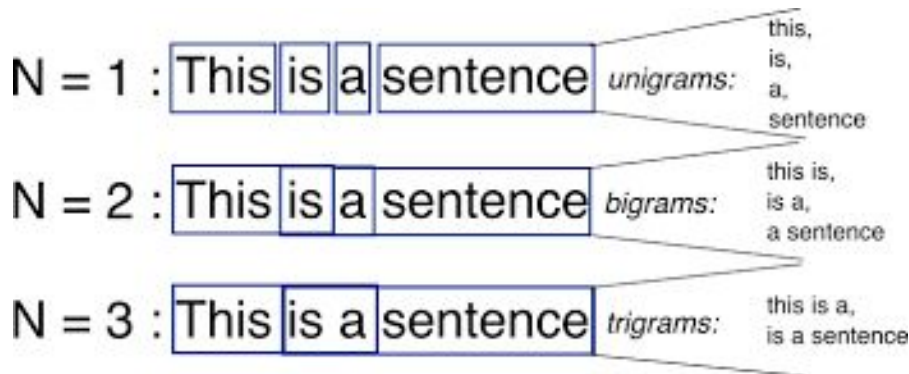


Puede crear un embedding de una palabra que nunca vió



*"Subsecuencia de N elementos de una secuencia dada"*

Otra forma de agrupar las palabras distinto a word2vec (N=1, unigram) en donde se busca aumentar el poder de generalización y a su vez poder hacer más variado el vocabulario.



## Character-level unigrams

| <u>Text</u> | <u>Token Sequence</u> | <u>Token Value</u> |
|-------------|-----------------------|--------------------|
| Dogs        | 1                     | D                  |
| Dogs        | 2                     | o                  |
| Dogs        | 3                     | g                  |
| Dogs        | 4                     | s                  |

## Character-level bigrams

| <u>Text</u> | <u>Token Sequence</u> | <u>Token Value</u> |
|-------------|-----------------------|--------------------|
| Dogs        | 1                     | Do                 |
| Dogs        | 2                     | og                 |
| Dogs        | 3                     | gs                 |

## Character-level trigrams

| <u>Text</u> | <u>Token Sequence</u> | <u>Token Value</u> |
|-------------|-----------------------|--------------------|
| Dogs        | 1                     | Dog                |
| Dogs        | 2                     | ogs                |

# N-gramas (por palabras)



Todas las hojas son del viento

Unigramas

“todas”

“las”

“hojas”

“son”

“del”

“viento”

Bigramas

“todas las”

“las hojas”

“hojas son”

“son del”

“del viento”

Trigramas

“Todas las hojas”

“las hojas son”

“hojas son del”

“son del viento”



# N-gramas (por caracteres)



Todas las hojas son del viento

Unigramas

"t"

"o"

"d"

"a"

"s"

"..."

Bigramas

"to"

"od"

"da"

"as"

"..."

Trigramas

"Tod"

"oda"

"das"

"as "

"..."

En esta idea se basa fastText

# Operaciones con Embeddings: tests de analogías



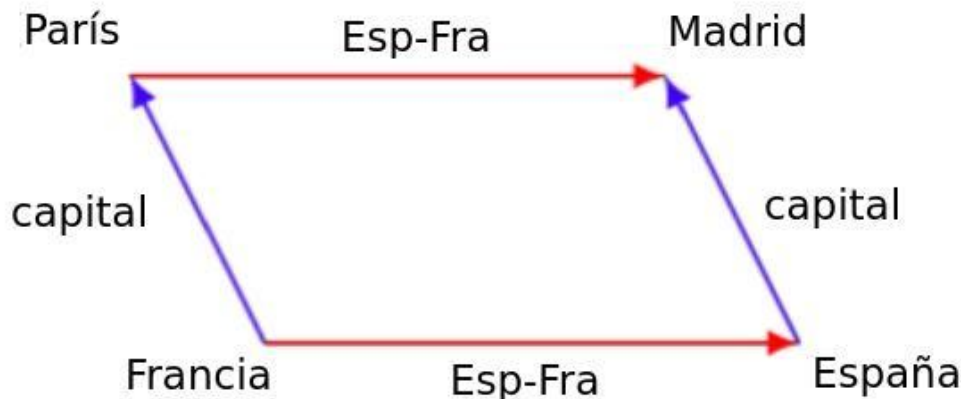
Una forma de testear la calidad de embeddings es probar su desempeño en tests de analogías:

París es a Francia lo que Madrid es a España.

Madrid y París corresponden a España y Francia

$$\overrightarrow{Paris} - \overrightarrow{Francia} \approx \overrightarrow{Madrid} - \overrightarrow{España}$$

$$\text{simcos}(\overrightarrow{Paris} - \overrightarrow{Francia}, \overrightarrow{Madrid} - \overrightarrow{España}) \approx 1$$





Colab 2.a





# Intervalo

# CBOW (Continuous Bag of Words)



## Adivinando la palabra por su contexto

El modelo recibe las palabras de alrededor (las "pistas") e intenta predecir cuál es la palabra que falta en el centro.

El \_\_\_\_\_ maúlla en el tejado.

### Características clave:

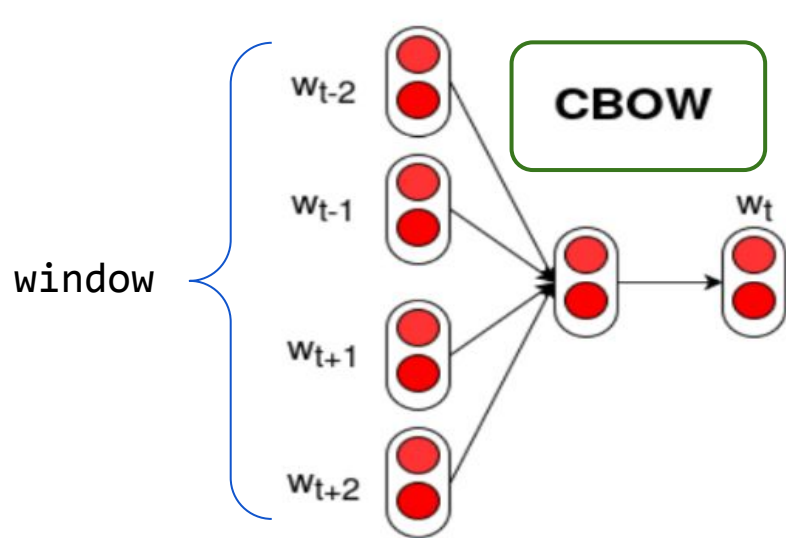
- **Muchos a Uno:** Varias palabras de entrada → Una palabra de salida.
- **Velocidad:** Es muy rápido de entrenar.
- **Uso ideal:** Funciona mejor cuando tenemos mucha cantidad de texto y queremos capturar el significado general de palabras muy comunes.



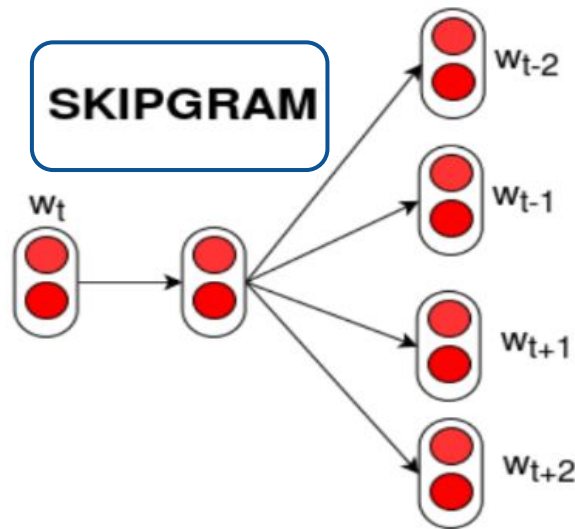
# ¿Cómo podemos crear nuestros word Embeddings?



Aprendiendo (con redes neuronales) vectores para cada palabra que maximicen la relación entre las palabras de contexto y la palabra objetivo. Esto es lo que se implementó en la librería **word2vec**.



$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_t | w_{t+j})$$



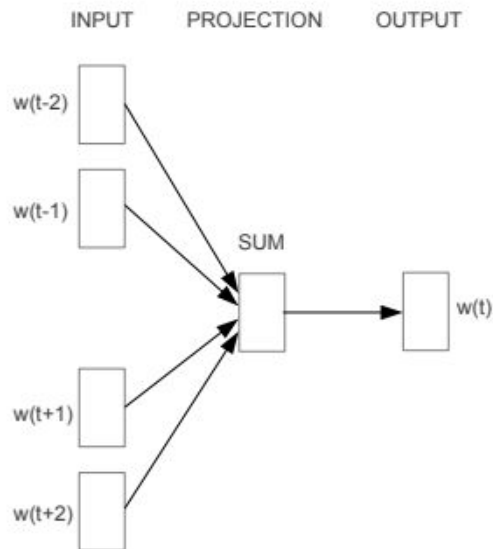
$$\frac{1}{T} \sum_{t=1}^T \sum_{-c \leq j \leq c, j \neq 0} \log p(w_{t+j} | w_t)$$

Se utiliza **OneHotEncoding** como representación del texto a la entrada del modelo

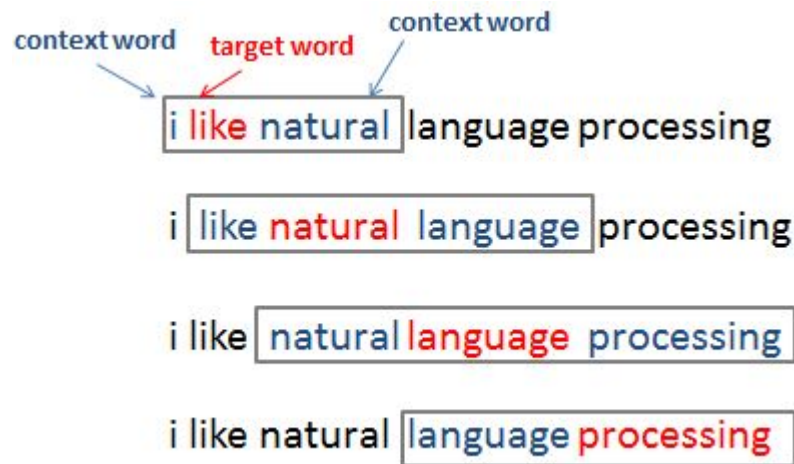
# Continuous Bag of Words Model (CBOW)



Utiliza como entrada el contexto de la palabra objetivo (palabras a izquierda y derecha de ella). El tamaño de la ventana determina cuántas palabras se tomarán para contextualizar el embedding.



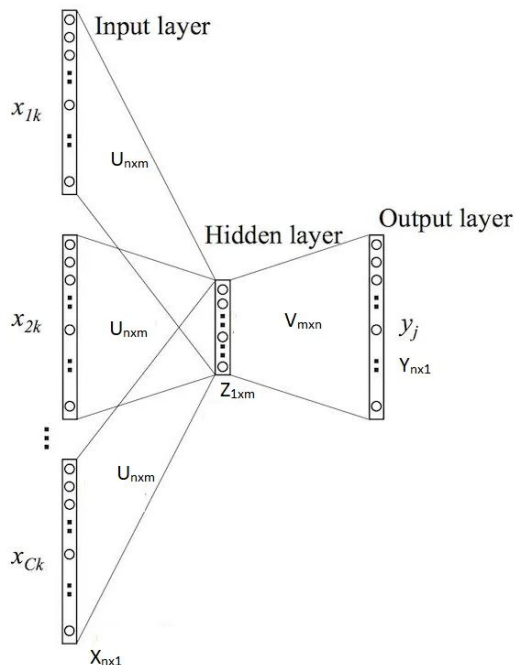
CBOW



# Continuous Bag of Words Model (CBOW)



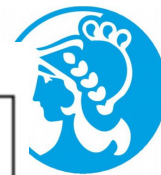
Utiliza como entrada el contexto de la palabra objetivo (palabras a izquierda y derecha de ella). El tamaño de la ventana determina cuántas palabras se tomarán para contextualizar el embedding.





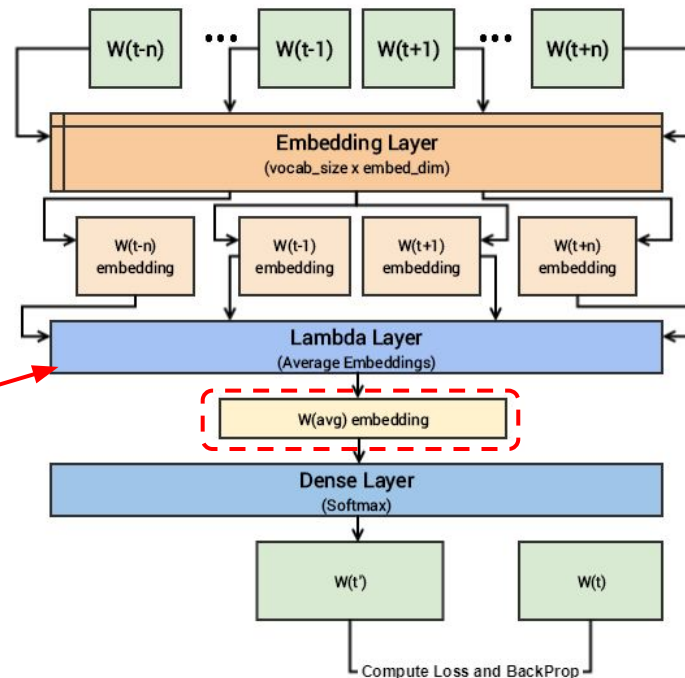
# CBOW - Entrenamiento

[LINK](#)



Para entrenar necesitamos tener el vocabulario del corpus y las sentencias organizadas por el tamaño de la ventana de entrada.

Los embeddings de cada palabra son el embedding promedio de todas las veces que se utilizó en el corpus.



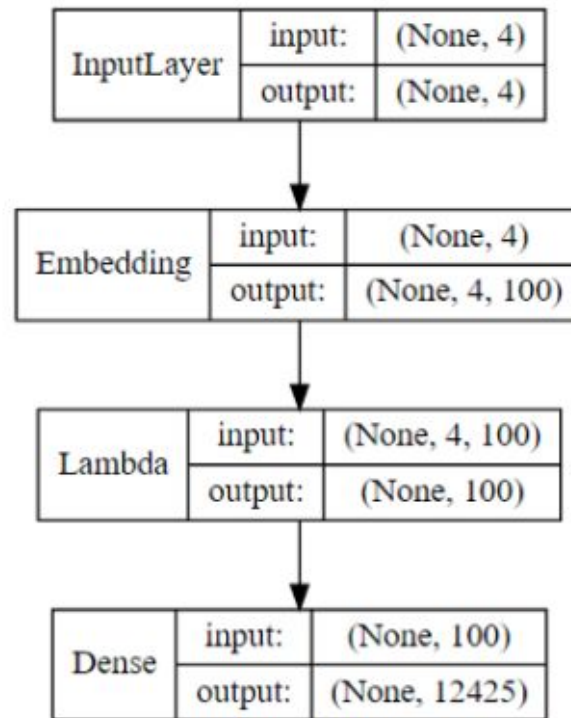
```
cbow = Sequential()
cbow.add(Embedding(input_dim=vocab_size, output_dim=embed_size, input_length=window_size*2))
cbow.add(Lambda(lambda x: K.mean(x, axis=1), output_shape=(embed_size,)))
cbow.add(Dense(vocab_size, activation='softmax'))
cbow.compile(loss='categorical_crossentropy', optimizer='rmsprop')
```

# CBOW - Entrenamiento



Con tan solo un corpus de 12425 palabras distintas y embedding de 100 dimensiones hay que entrenar **2.5 Millones de parámetros**

| Layer (type)                | Output Shape   | Param # |
|-----------------------------|----------------|---------|
| =====                       |                |         |
| embedding_1 (Embedding)     | (None, 4, 100) | 1242500 |
| =====                       |                |         |
| lambda_1 (Lambda)           | (None, 100)    | 0       |
| =====                       |                |         |
| dense_1 (Dense)             | (None, 12425)  | 1254925 |
| =====                       |                |         |
| Total params: 2,497,425     |                |         |
| Trainable params: 2,497,425 |                |         |
| Non-trainable params: 0     |                |         |



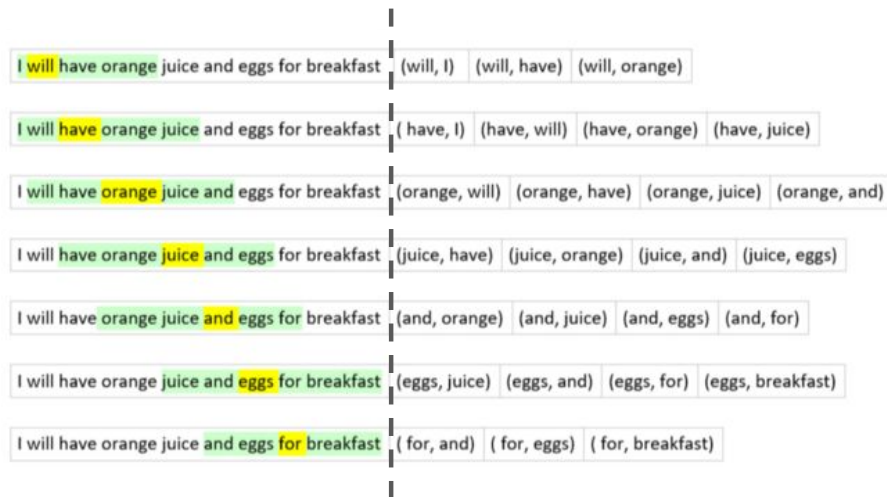


## Adivinando el contexto por una palabra

Al contrario de CBOW, este modelo intenta predecir las palabras que rodean (contexto) a una palabra objetivo. Se divide el output como pares [target, context]

### Características clave:

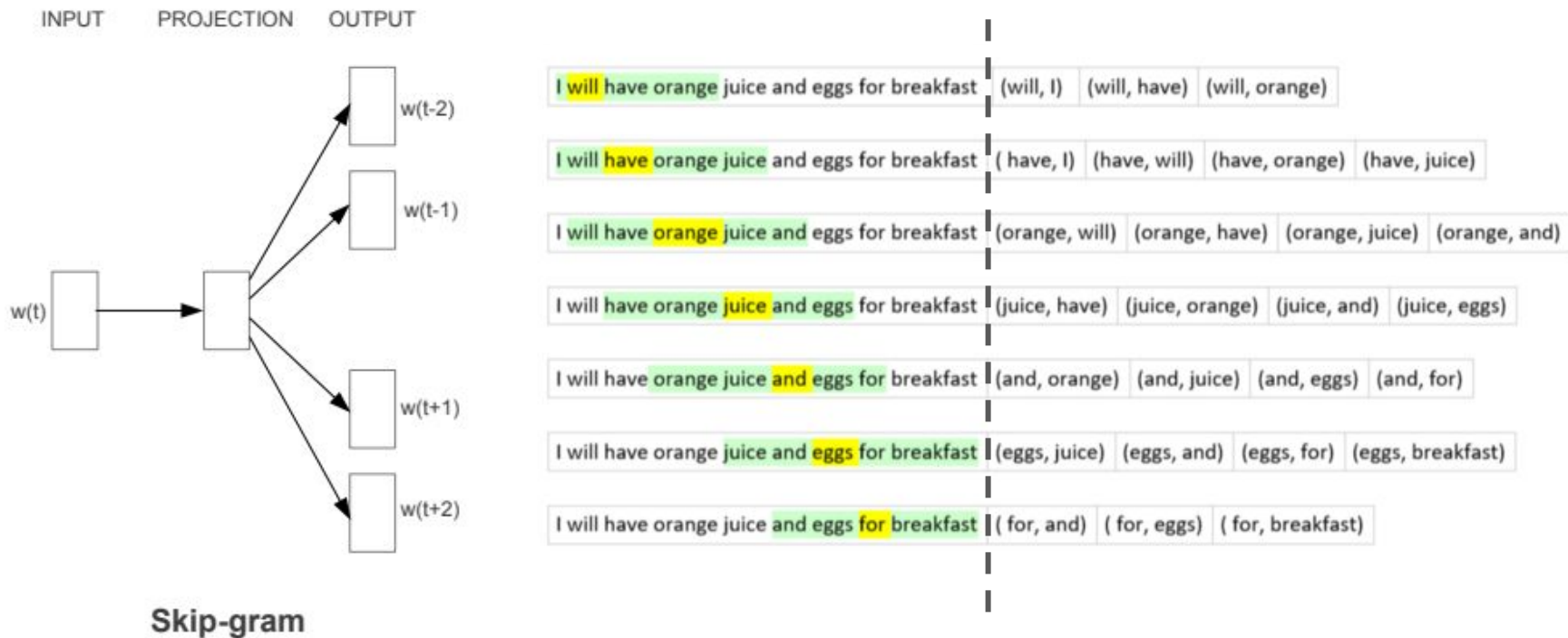
- **Uno a Muchos:** Una palabra de entrada → Varias palabras de contexto estimadas.
- **Precisión:** Es más lento que CBOW, pero mucho más preciso para palabras que no aparecen con frecuencia (palabras raras).
- **Uso ideal:** Es el preferido cuando el vocabulario es complejo o los datos no son gigantescos.



# Skip-Gram



Al contrario de CBOW, este modelo intenta predecir las palabras que rodean (contexto) a una palabra objetivo. Se divide el output como pares [target, context]



# Skip-Gram - Entrenamiento

[LINK](#)

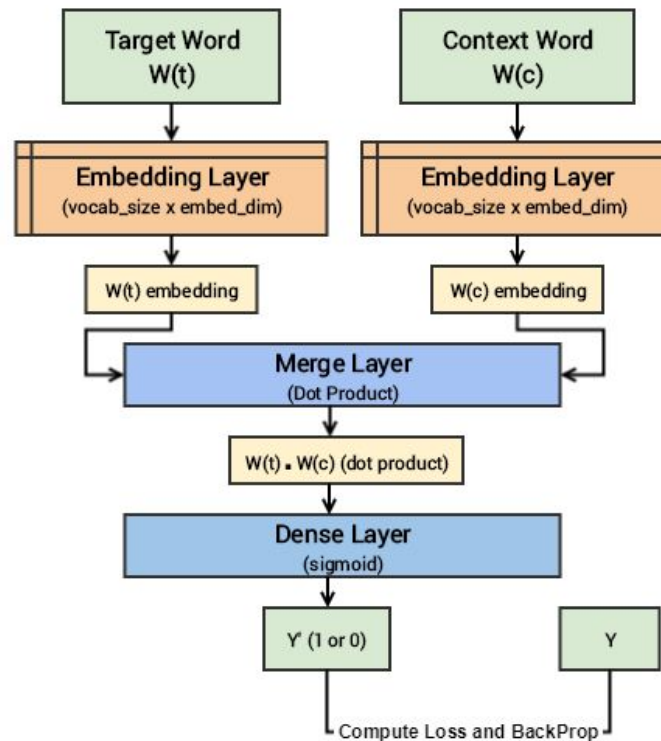


Por cada par [target, context] el sistema determina si las palabras tiene significado en contexto (1) o no lo tiene (0), buscando así acercar las palabras que tienen significado juntas (que se espera que estén juntas en el texto)

```
word_model = Sequential()
word_model.add(Embedding(vocab_size, embed_size,
                        embeddings_initializer="glorot_uniform",
                        input_length=1))
word_model.add(Reshape((embed_size, )))

context_model = Sequential()
context_model.add(Embedding(vocab_size, embed_size,
                           embeddings_initializer="glorot_uniform",
                           input_length=1))
context_model.add(Reshape((embed_size, )))

model = Sequential()
model.add(Merge([word_model, context_model], mode="dot"))
model.add(Dense(1, kernel_initializer="glorot_uniform", activation="sigmoid"))
```

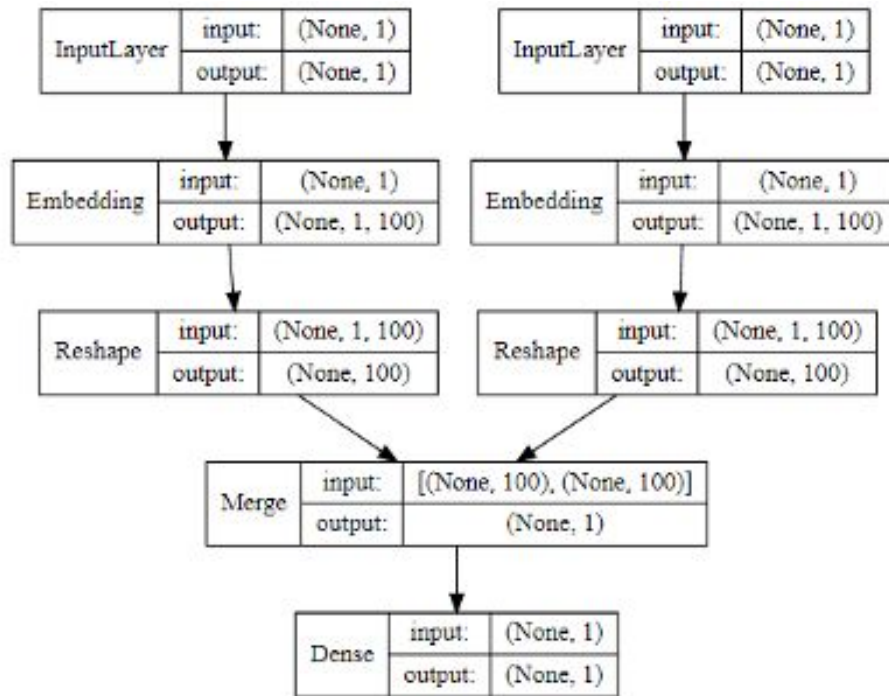


# Skip-Gram - Entrenamiento



Skip-Gram requiere más datos para lograr un buen resultado pero obtiene más información sobre el contexto del corpus en sus embeddings.

| Layer (type)                | Output Shape | Param # |
|-----------------------------|--------------|---------|
| merge_2 (Merge)             | (None, 1)    | 0       |
| dense_3 (Dense)             | (None, 1)    | 2       |
| Total params: 2,485,002     |              |         |
| Trainable params: 2,485,002 |              |         |
| Non-trainable params: 0     |              |         |





# Negative sampling

[LINK](#)



En SkipGram/CBOW la cantidad de parámetros a entrenar en la softmax es enorme:

**$Parametros = vocab\_size * embedding\_size \rightarrow \text{millones de parámetros}$**

Vanilla  
Skip-Gram

| W_output (old)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learning R. | grad_W_output | =     | W_output (new) |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|-------|----------------|--------|-------|--------|--------|--------|-------|--------|--------|-------|--------|-------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|--------|--------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|--------|--------|-------|--------|--------|--------|-------|--------|--------|--------|--------|-------|--------|-------|-------|-------|-------|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------|--------|--------|-------|--------|--------|--------|-------|--------|--------|-------|--------|-------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|--------|--------|
| <table><tr><td>-0.560</td><td>0.340</td><td>0.160</td></tr><tr><td>-0.910</td><td>-0.440</td><td>1.560</td></tr><tr><td>-1.210</td><td>-0.130</td><td>-1.320</td></tr><tr><td>1.670</td><td>-0.150</td><td>-1.030</td></tr><tr><td>1.720</td><td>-1.460</td><td>0.730</td></tr><tr><td>0.000</td><td>1.390</td><td>-0.120</td></tr><tr><td>-0.060</td><td>1.520</td><td>-0.790</td></tr><tr><td>0.800</td><td>1.850</td><td>-1.670</td></tr><tr><td>-1.370</td><td>1.320</td><td>-0.480</td></tr><tr><td>0.670</td><td>1.990</td><td>-1.850</td></tr><tr><td>-1.520</td><td>-1.740</td><td>-1.860</td></tr></table> | -0.560      | 0.340         | 0.160 | -0.910         | -0.440 | 1.560 | -1.210 | -0.130 | -1.320 | 1.670 | -0.150 | -1.030 | 1.720 | -1.460 | 0.730 | 0.000 | 1.390 | -0.120 | -0.060 | 1.520 | -0.790 | 0.800 | 1.850 | -1.670 | -1.370 | 1.320 | -0.480 | 0.670 | 1.990 | -1.850 | -1.520 | -1.740 | -1.860 | 0.05 | <table><tr><td>0.064</td><td>0.071</td><td>-0.014</td></tr><tr><td>0.098</td><td>0.015</td><td>0.063</td></tr><tr><td>0.069</td><td>0.089</td><td>0.045</td></tr><tr><td>0.014</td><td>0.085</td><td>0.079</td></tr><tr><td>-0.021</td><td>0.067</td><td>0.071</td></tr><tr><td>-0.098</td><td>-0.088</td><td>0.091</td></tr><tr><td>-0.072</td><td>-0.078</td><td>-0.089</td></tr><tr><td>0.046</td><td>-0.079</td><td>-0.053</td></tr><tr><td>-0.049</td><td>-0.087</td><td>0.025</td></tr><tr><td>-0.060</td><td>0.092</td><td>0.042</td></tr><tr><td>0.074</td><td>0.050</td><td>0.070</td></tr></table> | 0.064 | 0.071 | -0.014 | 0.098 | 0.015 | 0.063 | 0.069 | 0.089 | 0.045 | 0.014 | 0.085 | 0.079 | -0.021 | 0.067 | 0.071 | -0.098 | -0.088 | 0.091 | -0.072 | -0.078 | -0.089 | 0.046 | -0.079 | -0.053 | -0.049 | -0.087 | 0.025 | -0.060 | 0.092 | 0.042 | 0.074 | 0.050 | 0.070 |  | <table><tr><td>-0.563</td><td>0.336</td><td>0.161</td></tr><tr><td>-0.915</td><td>-0.441</td><td>1.557</td></tr><tr><td>-1.213</td><td>-0.134</td><td>-1.322</td></tr><tr><td>1.669</td><td>-0.154</td><td>-1.034</td></tr><tr><td>1.721</td><td>-1.463</td><td>0.726</td></tr><tr><td>0.005</td><td>1.394</td><td>-0.125</td></tr><tr><td>-0.056</td><td>1.524</td><td>-0.786</td></tr><tr><td>0.798</td><td>1.854</td><td>-1.667</td></tr><tr><td>-1.368</td><td>1.324</td><td>-0.481</td></tr><tr><td>0.673</td><td>1.985</td><td>-1.852</td></tr><tr><td>-1.524</td><td>-1.743</td><td>-1.864</td></tr></table> | -0.563 | 0.336 | 0.161 | -0.915 | -0.441 | 1.557 | -1.213 | -0.134 | -1.322 | 1.669 | -0.154 | -1.034 | 1.721 | -1.463 | 0.726 | 0.005 | 1.394 | -0.125 | -0.056 | 1.524 | -0.786 | 0.798 | 1.854 | -1.667 | -1.368 | 1.324 | -0.481 | 0.673 | 1.985 | -1.852 | -1.524 | -1.743 | -1.864 |
| -0.560                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.340       | 0.160         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.910                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.440      | 1.560         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.210                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.130      | -1.320        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 1.670                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.150      | -1.030        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 1.720                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.460      | 0.730         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.390       | -0.120        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.060                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.520       | -0.790        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.850       | -1.670        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.320       | -0.480        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.670                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.990       | -1.850        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.520                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -1.740      | -1.860        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.064                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.071       | -0.014        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.098                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.015       | 0.063         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.069                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.089       | 0.045         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.014                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.085       | 0.079         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.067       | 0.071         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.098                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.088      | 0.091         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.072                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.078      | -0.089        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.046                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.079      | -0.053        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.049                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.087      | 0.025         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.060                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.092       | 0.042         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.074                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.050       | 0.070         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.563                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.336       | 0.161         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.915                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.441      | 1.557         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.213                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.134      | -1.322        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 1.669                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.154      | -1.034        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 1.721                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.463      | 0.726         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.394       | -0.125        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -0.056                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.524       | -0.786        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.798                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.854       | -1.667        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.368                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.324       | -0.481        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| 0.673                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.985       | -1.852        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| -1.524                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -1.743      | -1.864        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |
| (11X3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             | (11X3)        |       | (11X3)         |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |       |        |       |       |       |       |       |       |       |       |       |        |       |       |        |        |       |        |        |        |       |        |        |        |        |       |        |       |       |       |       |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |

Negative  
Sampling

| W_output (old)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Learning R.   | grad_W_output | =     | W_output (new) |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-------|----------------|--------|-------|--------|--------|--------|-------|--------|--------|-------|--------|-------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|-------|--------|-------|-------|--------|--------|--------|--------|-------------------|--------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------|--------|--------|-------|--------|--------|--------|-------|--------|--------|-------|--------|-------|-------|-------|--------|--------|-------|--------|--------------|--------------|---------------|---------------|--------------|---------------|--------------|--------------|---------------|---------------|---------------|---------------|
| <table><tr><td>-0.560</td><td>0.340</td><td>0.160</td></tr><tr><td>-0.910</td><td>-0.440</td><td>1.560</td></tr><tr><td>-1.210</td><td>-0.130</td><td>-1.320</td></tr><tr><td>1.670</td><td>-0.150</td><td>-1.030</td></tr><tr><td>1.720</td><td>-1.460</td><td>0.730</td></tr><tr><td>0.000</td><td>1.390</td><td>-0.120</td></tr><tr><td>-0.060</td><td>1.520</td><td>-0.790</td></tr><tr><td>0.800</td><td>1.850</td><td>-1.670</td></tr><tr><td>-1.370</td><td>1.320</td><td>-0.480</td></tr><tr><td>0.670</td><td>1.990</td><td>-1.850</td></tr><tr><td>-1.520</td><td>-1.740</td><td>-1.860</td></tr></table> | -0.560        | 0.340         | 0.160 | -0.910         | -0.440 | 1.560 | -1.210 | -0.130 | -1.320 | 1.670 | -0.150 | -1.030 | 1.720 | -1.460 | 0.730 | 0.000 | 1.390 | -0.120 | -0.060 | 1.520 | -0.790 | 0.800 | 1.850 | -1.670 | -1.370 | 1.320 | -0.480 | 0.670 | 1.990 | -1.850 | -1.520 | -1.740 | -1.860 | <div>0.05</div> × | <div>Not computed!</div> |  | <table><tr><td>-0.560</td><td>0.340</td><td>0.160</td></tr><tr><td>-0.910</td><td>-0.440</td><td>1.560</td></tr><tr><td>-1.210</td><td>-0.130</td><td>-1.320</td></tr><tr><td>1.670</td><td>-0.150</td><td>-1.030</td></tr><tr><td>1.720</td><td>-1.460</td><td>0.730</td></tr><tr><td>0.000</td><td>1.390</td><td>-0.120</td></tr><tr><td>-0.060</td><td>1.520</td><td>-0.790</td></tr><tr><td><b>0.798</b></td><td><b>1.849</b></td><td><b>-1.672</b></td></tr><tr><td><b>-1.366</b></td><td><b>1.318</b></td><td><b>-0.477</b></td></tr><tr><td><b>0.667</b></td><td><b>1.985</b></td><td><b>-1.847</b></td></tr><tr><td><b>-1.523</b></td><td><b>-1.744</b></td><td><b>-1.858</b></td></tr></table> | -0.560 | 0.340 | 0.160 | -0.910 | -0.440 | 1.560 | -1.210 | -0.130 | -1.320 | 1.670 | -0.150 | -1.030 | 1.720 | -1.460 | 0.730 | 0.000 | 1.390 | -0.120 | -0.060 | 1.520 | -0.790 | <b>0.798</b> | <b>1.849</b> | <b>-1.672</b> | <b>-1.366</b> | <b>1.318</b> | <b>-0.477</b> | <b>0.667</b> | <b>1.985</b> | <b>-1.847</b> | <b>-1.523</b> | <b>-1.744</b> | <b>-1.858</b> |
| -0.560                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.340         | 0.160         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -0.910                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.440        | 1.560         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -1.210                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.130        | -1.320        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 1.670                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.150        | -1.030        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 1.720                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.460        | 0.730         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.390         | -0.120        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -0.060                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.520         | -0.790        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 0.800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.850         | -1.670        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -1.370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.320         | -0.480        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 0.670                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.990         | -1.850        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -1.520                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -1.740        | -1.860        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -0.560                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.340         | 0.160         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -0.910                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.440        | 1.560         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -1.210                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -0.130        | -1.320        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 1.670                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.150        | -1.030        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 1.720                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.460        | 0.730         |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| 0.000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.390         | -0.120        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| -0.060                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1.520         | -0.790        |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| <b>0.798</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>1.849</b>  | <b>-1.672</b> |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| <b>-1.366</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>1.318</b>  | <b>-0.477</b> |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| <b>0.667</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>1.985</b>  | <b>-1.847</b> |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| <b>-1.523</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>-1.744</b> | <b>-1.858</b> |       |                |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |
| (11X3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               | (11X3)        |       | (11X3)         |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |       |       |        |        |       |        |       |       |        |        |        |        |                   |                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |       |       |        |        |       |        |        |        |       |        |        |       |        |       |       |       |        |        |       |        |              |              |               |               |              |               |              |              |               |               |               |               |

Positive sample, w\_o

|       |       |       |
|-------|-------|-------|
| 0.031 | 0.030 | 0.041 |
|-------|-------|-------|

Negative sample, k=1

|        |       |        |
|--------|-------|--------|
| -0.090 | 0.031 | -0.065 |
|--------|-------|--------|

Negative sample, k=2

|       |       |        |
|-------|-------|--------|
| 0.056 | 0.098 | -0.061 |
|-------|-------|--------|

Negative sample, k=3

|       |       |        |
|-------|-------|--------|
| 0.069 | 0.084 | -0.044 |
|-------|-------|--------|

En cada iteración se observa la palabras [target, contexto] y "K" palabras aleatorias del corpus.

El objetivo es optimizar cómputo aproximando la softmax. Además funciona como regularización. Para corpus pequeños, el muestreo debe ser mayor.

# Visualizar embeddings en baja dimensionalidad:

[LINK](#)



## t-SNE (t-distributed stochastic neighbor embedding)



Técnica de reducción de dimensionalidad no-lineal (a diferencia de PCA).



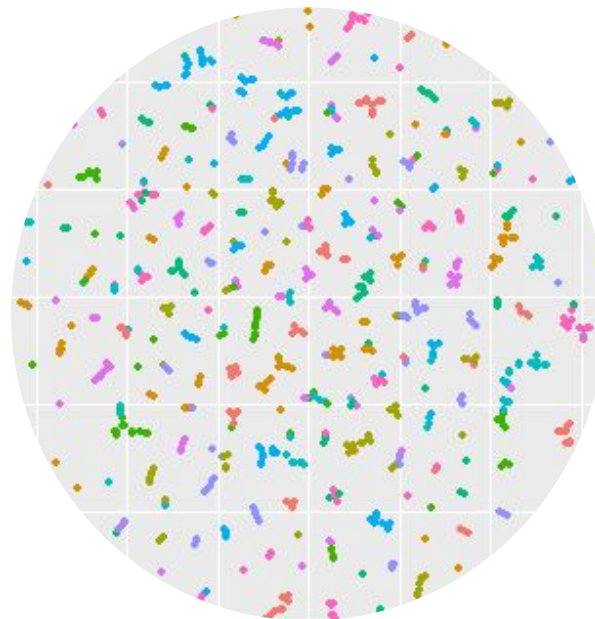
Intenta reproducir en baja dimensionalidad, la localidad de los datos en alta dimensionalidad.



Es estocástica, a priori los resultados no se repiten.



Por su carácter estocástico es sólo recomendable como herramienta de visualización y exploratoria.







## Intervalo

# Gensim - Doc2Vec paragraph embeddings

[LINK](#)



Utilizaremos esta librería que nos facilita generar embeddings tipo Skip-Gram o CBOW de nuestros corpus



- Librería de Python
- Existe desde 2009 y nació originalmente para topic modelling
- Muy popular y muy simple de utilizar



Colab 2.b





Crear sus propios vectores con Gensim basado en lo visto en clase con otro dataset.

Probar términos de interés y explicar similitudes en el espacio de embeddings.

Intentar plantear y probar tests de analogías. Graficar los embeddings resultantes.

Sacar conclusiones.



# Algunos recursos para descargar corpora de texto



[Project Gutenberg](#)

Compilación de literatura completa de dominio público principalmente en inglés.

[Textos.info](#)

Compilación de literatura completa de dominio público en español.



Google  
BERT

**¿Qué cambió?** El modelo ya no lee palabra por palabra de forma aislada. Mira la frase **entera**.

**Innovación:** El "Embedding" ya no se busca en una tabla fija; se **calcula en tiempo real** dependiendo de las palabras vecinas.



## Sentence embeddings: ¿Cómo representamos una idea completa?

**No es solo sumar palabras:** No basta con sumar los vectores de "No", "es", "bueno". El orden y la estructura importan.

**¿Qué son?** Son vectores que representan el significado global de una oración o un párrafo entero.

## Embeddings multimodales



¿Y si usamos el **mismo mapa para palabras y para fotos**?

**¿Cómo funciona?** El modelo se entrena con millones de imágenes y sus leyendas. Aprende que el vector de la foto de un perro debe estar en el mismo lugar que el vector de la palabra "perro".



## Recomendaciones

## Buscadores

## Base de LLMs



Gemini

